SPT Progress Meeting

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Omar Mohamed – Practical Work (GBST-BERT)

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Recap

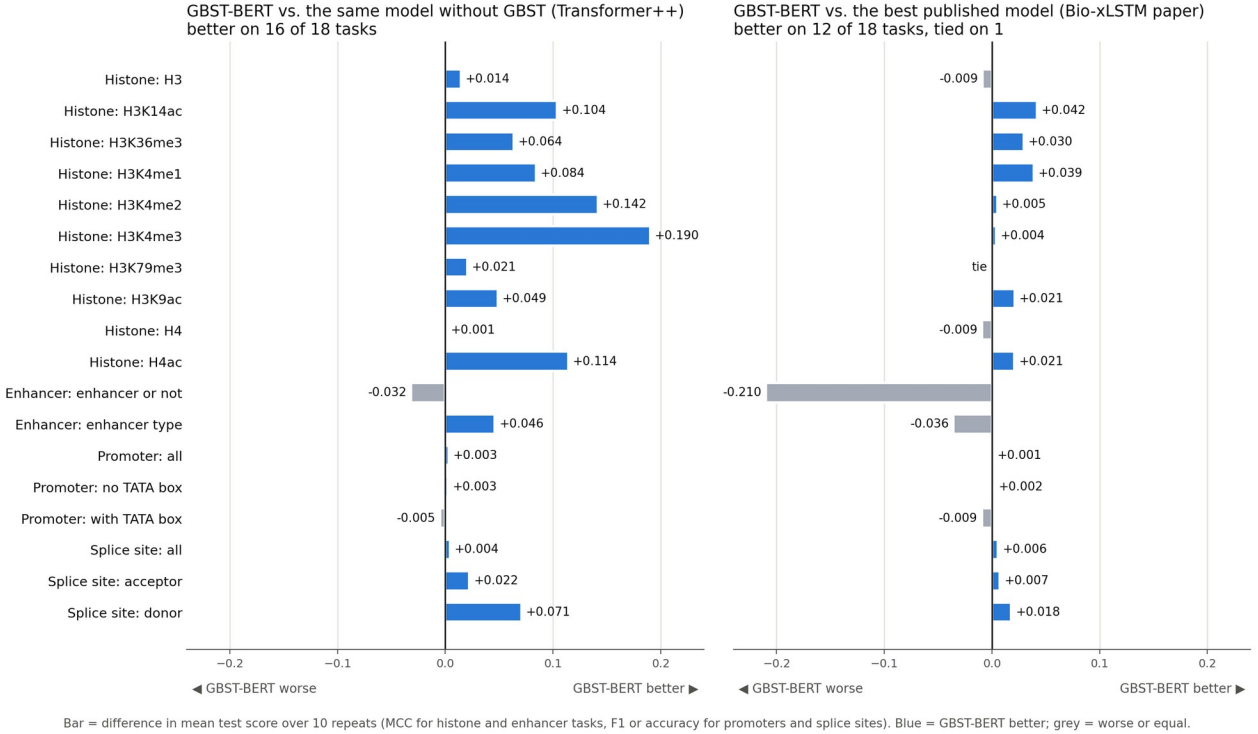
- **Status at the last meeting:** GBST-BERT = learned tokenizer (GBST) + small BERT encoder, 2.1M parameters
  - Pretrained on the human genome (hg38) with the DNA-xLSTM short-context protocol: 10,000 steps
  - Compared with the same model without GBST (Transformer++, 2.0M parameters)
- **Next step decided:** test the model on real biology tasks
  - Nucleotide Transformer benchmark (18 tasks), evaluated exactly like the Bio-xLSTM paper

Progress (1/3): pretraining and benchmark

- **Pretraining (test set):** masked-base accuracy 56.44% vs 56.33% for Transformer++ (loss 0.975 vs 0.977)
- **Nucleotide Transformer benchmark:** 18 tasks × 10 repeats, Bio-xLSTM protocol
  - Histone marks, enhancers, promoters, splice sites
  - Paper's learning rate and batch size per task (Table A5); best validation epoch tested
- **Result:** best of all 7 models on 13 of 18 tasks
  - Better than Transformer++ on 16 of 18 tasks → the learned tokenizer helps
  - Better than the best published model on 12 tasks, tied on 1
  - Weakest task: enhancer or not (−0.21 vs best paper model)

Progress (2/3): GBST-BERT vs other models

- Blue = GBST-BERT better
- Left: same model without GBST
- Right: best model in the Bio-xLSTM paper
- Score = mean test score over 10 repeats



Progress (3/3): findings and questions

- **What GBST learned:** 98% of its weight on 1-base blocks, so it hardly merges bases
  - The gain probably comes from the extra convolution (local context), not from real "words"
- **Implemented / written:** one clean model file (gbst_bert.py, tested against the training code)
  - Full notebook that re-runs every step on the server; report, poster, slides; all on GitHub
- **Questions for you:**
  - Is this result enough for the practical work, or should I do more experiments?
  - Learning-rate search per task, or keep the paper's settings for a fair comparison?
  - Longer pretraining (20k steps) or longer sequences (2048)?

JKU

Next steps

- **Before the next meeting (depending on your answers):**
  - Small learning-rate / batch-size search on the weakest tasks (enhancers)
  - Check whether GBST merges more bases with longer pretraining or longer sequences
  - Finish the written report with the benchmark results
- Keep this deck updated with Recap / Progress / Next steps for every meeting